



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 3310

Digital Fundamentals and System Design

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Outline Lecture 7 FSM – Design & Implementation



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1 Learning Outcomes

2 FSM Design Steps

3 Mealy vs. Moore

4 Example: Serial Adder

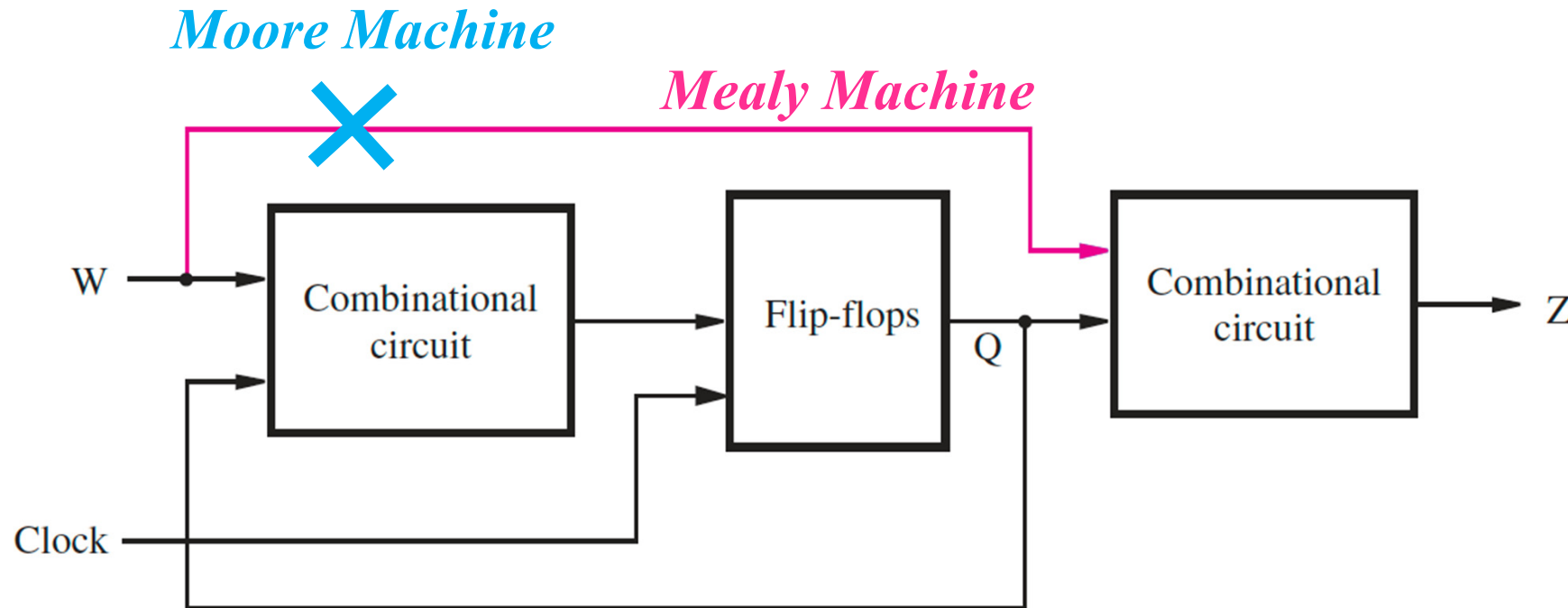
Main Content

1. Learning Outcomes
2. FSM Types

1. Learning Outcomes

1. Describe two distinct types of FSMs
2. Describe the **Mealy** and **Moore** Machines
3. Describe design steps in a general sequential circuit
4. Design a two-bit **Mealy** Machine sequence detector
5. Design a two-bit **Moore** Machine sequence detector
6. Build a state diagram for a **Mealy** Serial Adder
7. Convert a **Mealy** state diagram to a **Moore** state diagram (Serial Adder)
8. Convert a **Moore** state diagram to a **Mealy** state diagram (Serial Adder)

2. General Form of a Sequential Circuit

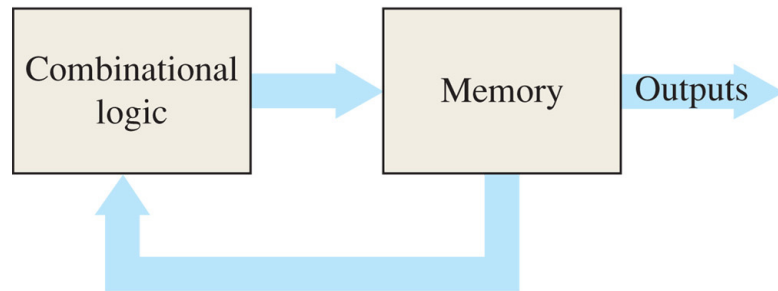


2. FSM Types

Two types of sequential logic (FSMs).

Moore Machine:

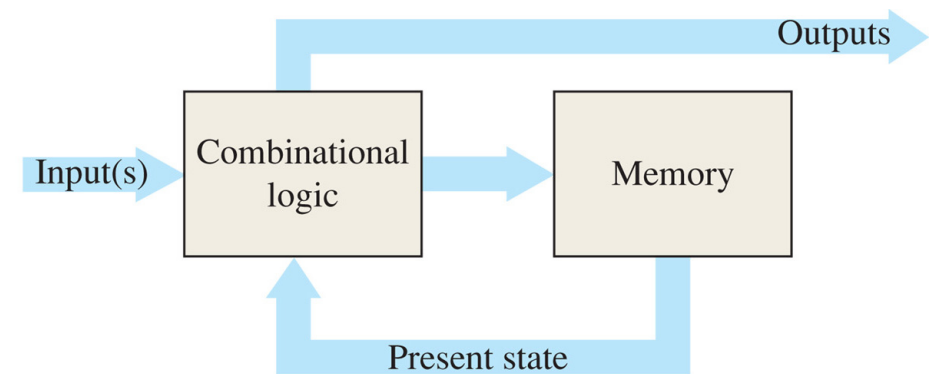
FSM output is **synchronous** with CLOCK
(output CANNOT change WITHOUT clock edge)



(a) Moore machine

Mealy Machine:

FSM output is **asynchronous** with CLOCK
(output CAN change with inputs any time)



(b) Mealy machine

1. Obtain specification of the desired circuit.
 2. Create a state diagram from specification.
 3. Create a state table from state diagram.
 4. Perform state minimization.
 5. Perform state assignment.
 6. Derive the next-state and output logic expressions.
 7. Implement circuit described by logic.
- Steps 4 & 5 are not in counter design procedure*

Example (Moore)

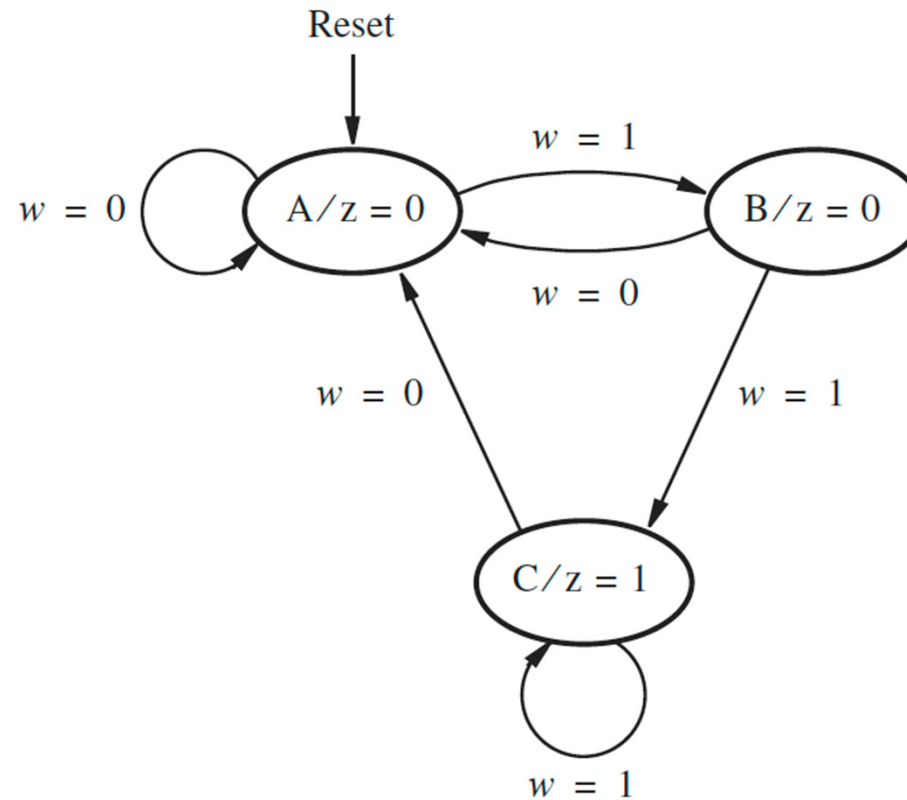
- Circuit has one input, w , and one output, z .
- Changes occur on positive clock edge.
- z is 1 if w is 1 during last two clock cycles.

“The specification requires that the output z be equal to 1 in the clock cycle that follows the detection of the second occurrence of $w = 1$ ”

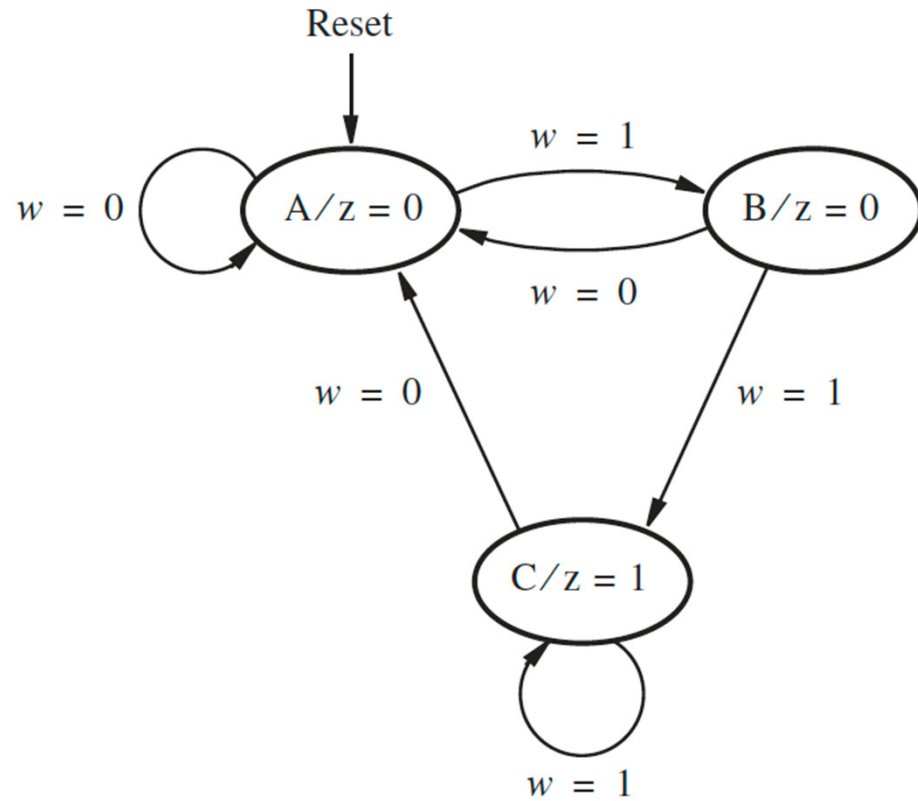
Sequences of input and output signals

Clockcycle:	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
w :	0	1	0	1	1	0	1	1	1	0	1
z :	0	0	0	0	0	1	0	0	1	1	0

1. Create a state diagram from specification



2. Create a state table from state diagram



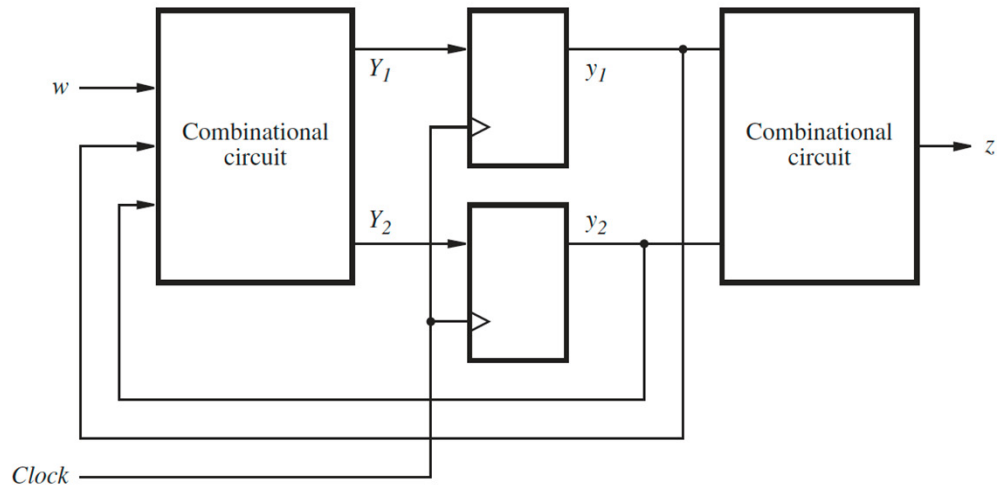
State table

Present state	Next state		Output z
	$w = 0$	$w = 1$	
A	A	B	0
B	A	C	0
C	A	C	1

From diagram to table

3. Perform state minimization (no need; already minimum) .

4. Perform state assignment.



Present state	Next state		Output z
	$w = 0$	$w = 1$	
A	A	B	0
B	A	C	0
C	A	C	1

	Present state $y_2 y_1$	Next state		Output z
		$w = 0$	$w = 1$	
		$Y_2 Y_1$	$Y_2 Y_1$	
A	00	00	01	0
B	01	00	10	0
C	10	00	10	1
	11	<i>dd</i>	<i>dd</i>	<i>d</i>

From diagram to table



5. Derive the next-state and output logic expressions

	Present state y_2y_1	Next state		Output z
		$w = 0$	$w = 1$	
		Y_2Y_1	Y_2Y_1	
A	00	00	01	0
B	01	00	10	0
C	10	00	10	1
	11	dd	dd	d

w	y_2y_1	00	01	11	10
		0	0	d	0
1		1	0	d	0

Ignoring don't cares

$$Y_1 = w\bar{y}_1\bar{y}_2$$

Using don't cares

$$Y_1 = w\bar{y}_1\bar{y}_2$$

w	y_2y_1	00	01	11	10
		0	0	d	0
1		0	1	d	1

$$Y_2 = wy_1\bar{y}_2 + w\bar{y}_1y_2$$

$$\begin{aligned} Y_2 &= wy_1 + wy_2 \\ &= w(y_1 + y_2) \end{aligned}$$

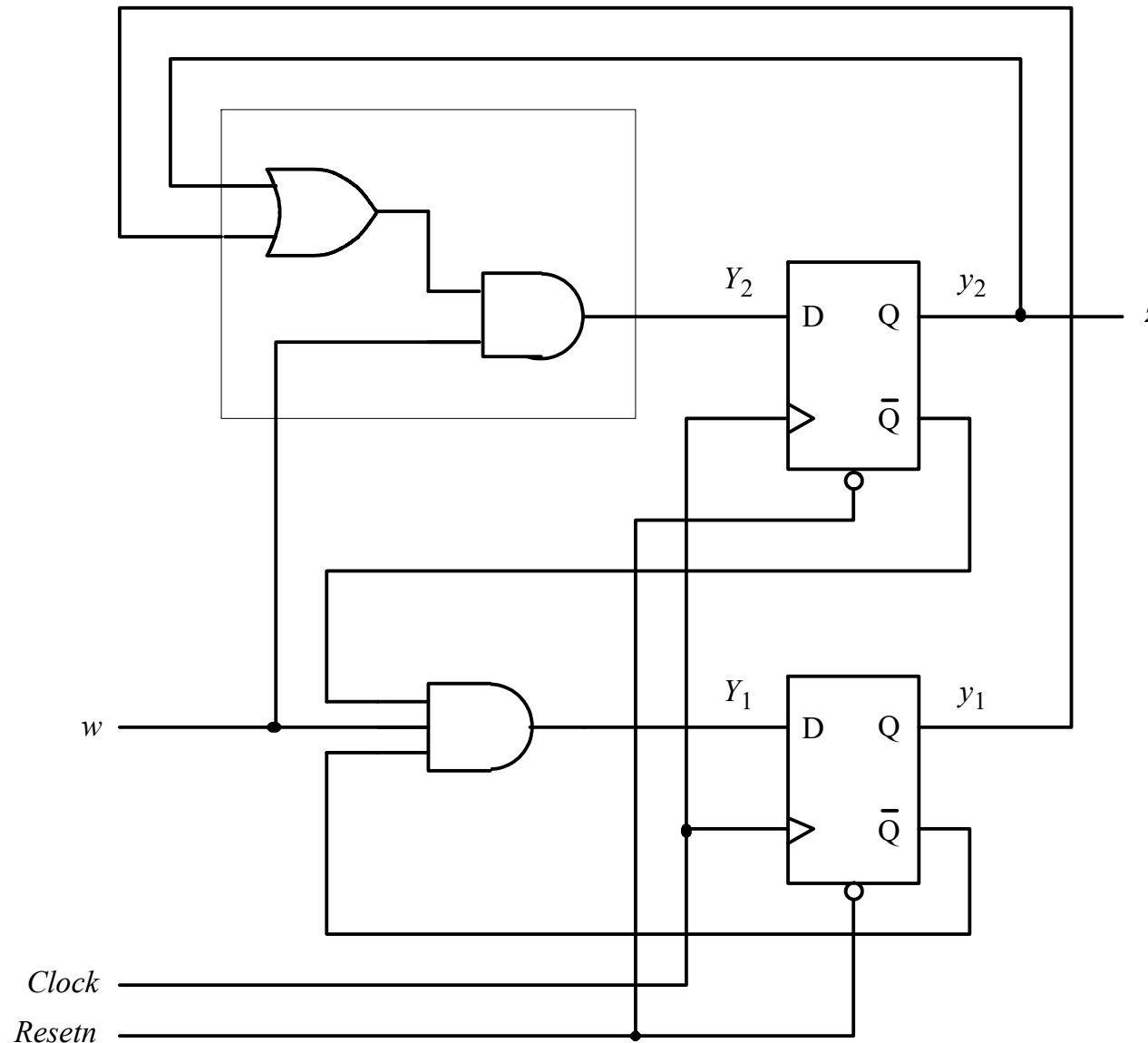
y_2	y_1	0	1
		0	0
1		1	d

$$z = \bar{y}_1y_2$$

$$z = y_2$$

From diagram to table

6. Implement circuit described by logic.



Using don't cares

$$Y_1 = w\bar{y}_1\bar{y}_2$$

$$\begin{aligned} Y_2 &= wy_1 + wy_2 \\ &= w(y_1 + y_2) \end{aligned}$$

$$z = y_2$$

From diagram to table

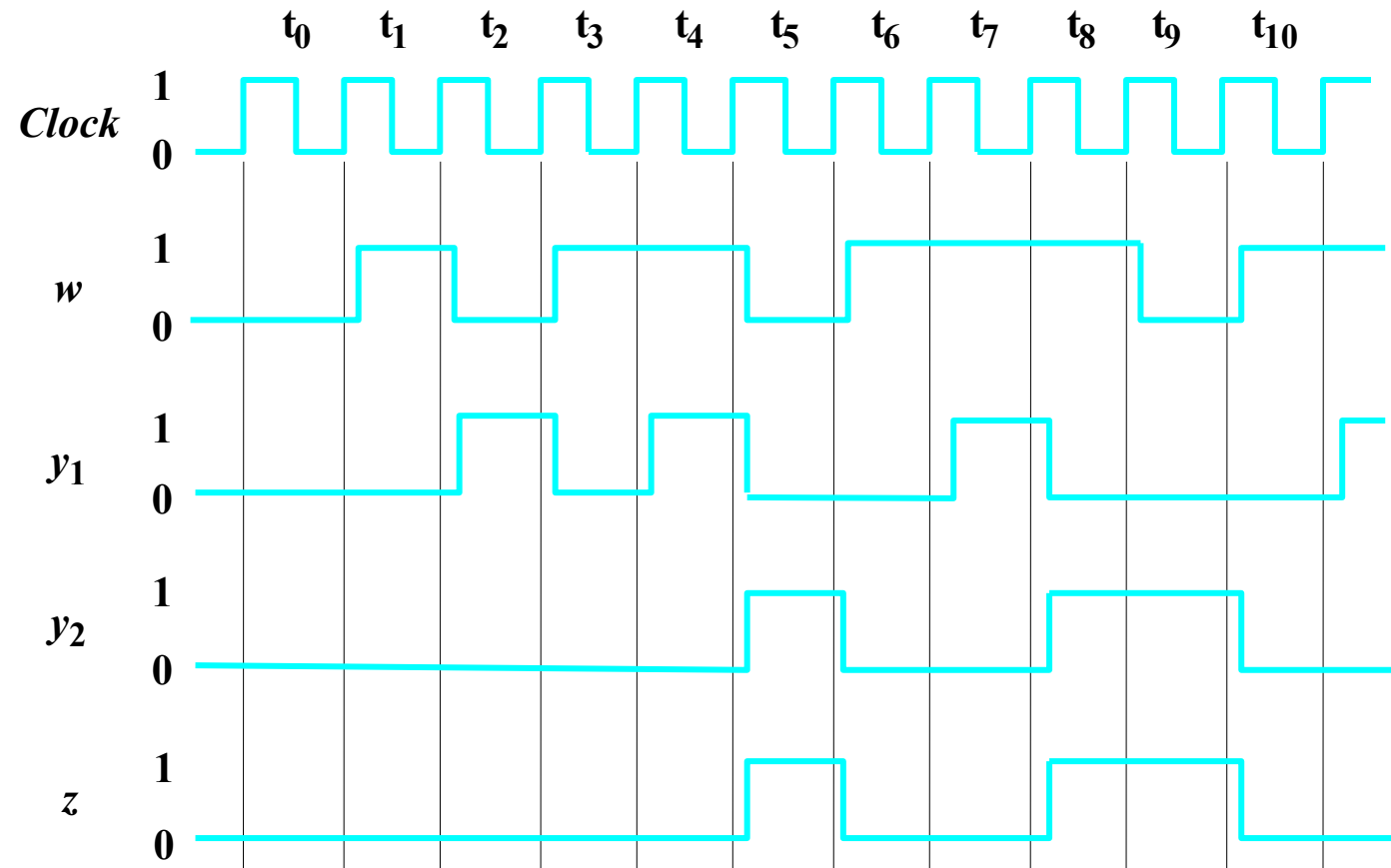
Using don't cares

$$Y_1 = w\bar{y}_1\bar{y}_2$$

$$\begin{aligned} Y_2 &= wy_1 + wy_2 \\ &= w(y_1 + y_2) \end{aligned}$$

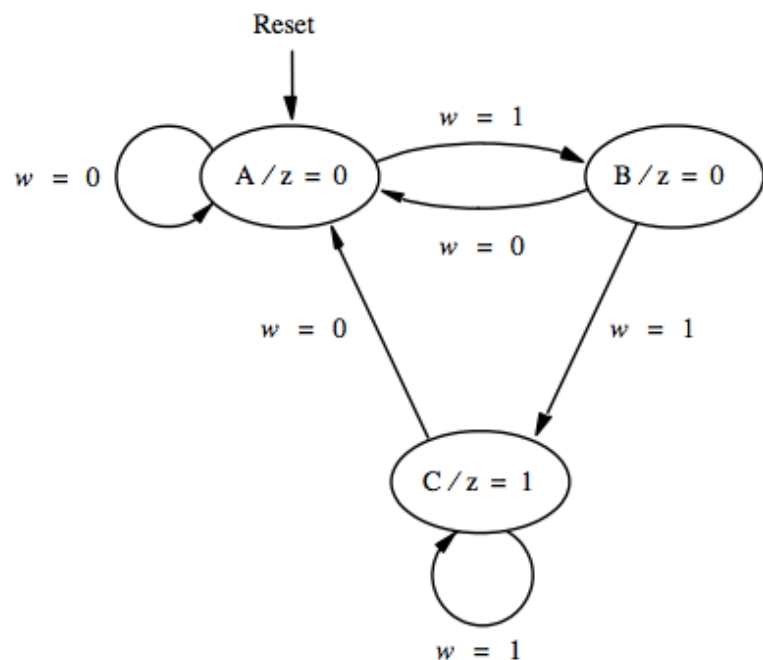
$$z = y_2$$

Clock cycle:	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
w :	0	1	0	1	1	0	1	1	1	0	1
z :	0	0	0	0	0	1	0	0	1	1	0





From diagram to table



	Present state	Next state		Output z
		$w = 0$	$w = 1$	
	$y_2 y_1$	$Y_2 Y_1$	$Y_2 Y_1$	
A	00	00	01	0
B	01	00	11	0
C	11	00	11	1
	10	dd	dd	d

Present state	Next state		Output z
	$w = 0$	$w = 1$	
A	A	B	0
B	A	C	0
C	A	C	1

	Present state	Next state		Output z
		$w = 0$	$w = 1$	
	$y_2 y_1$	$Y_2 Y_1$	$Y_2 Y_1$	
A	00	00	01	0
B	01	00	10	0
C	10	00	10	1
	11	dd	dd	d

Slightly different State Assignment??

From diagram to table



	Present state	Next state		Output z
		$w = 0$	$w = 1$	
	$y_2 y_1$	$Y_2 Y_1$	$Y_2 Y_1$	
A	00	00	01	0
B	01	00	11	0
C	11	00	11	1
	10	dd	dd	d

$y_2 y_1$					
w		00	01	11	10
		0	0	0	d
1		1	1	1	d

$y_2 y_1$					
w		00	01	11	10
		0	0	0	d
1		0	1	1	d

$y_2 y_1$			
y_2		0	1
		0	0
1		d	1

$$Y_1 = w$$

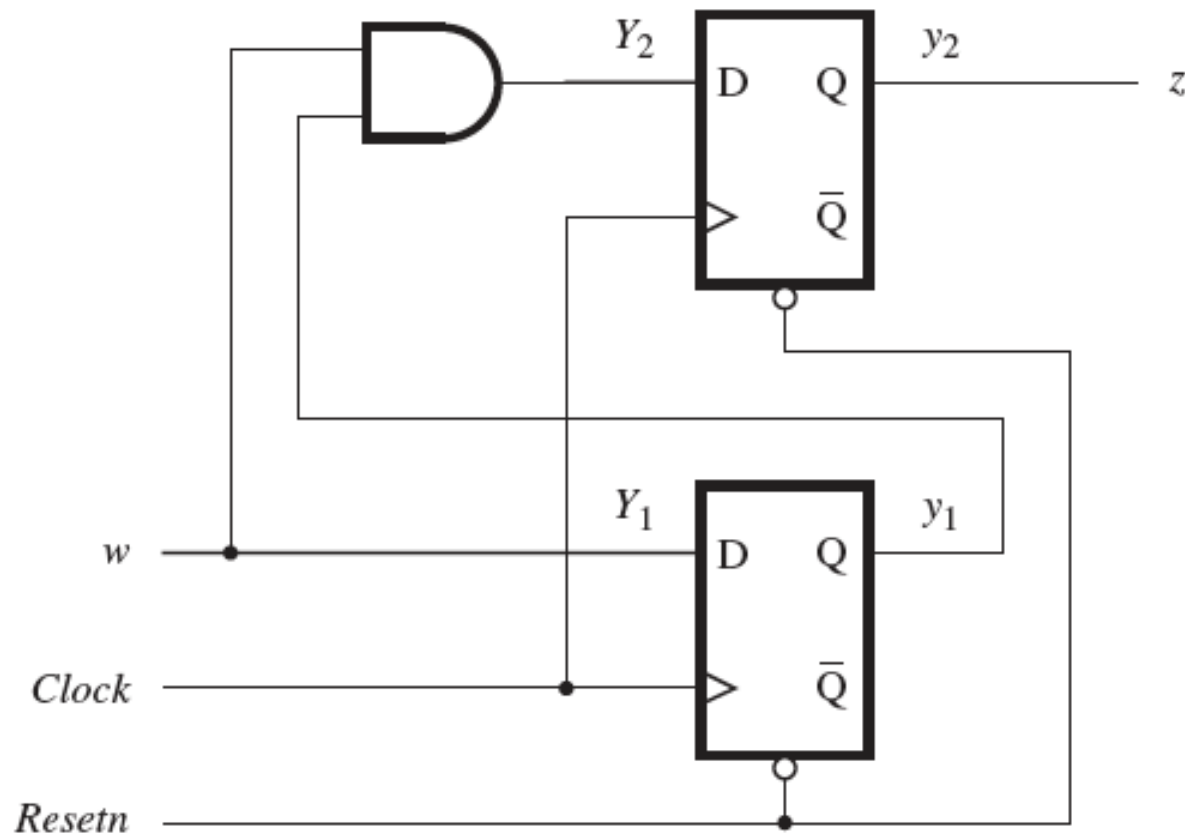
$$Y_2 = w y_1$$

$$z = y_2$$

From diagram to table



Using don't cares



$$Y_1 = w$$

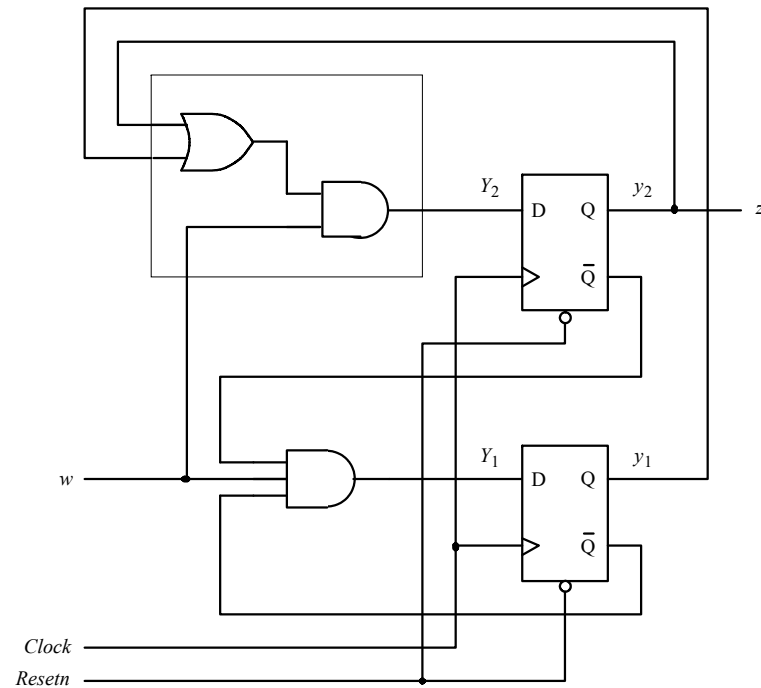
$$Y_2 = wy_1$$

$$z = y_2$$

From diagram to table

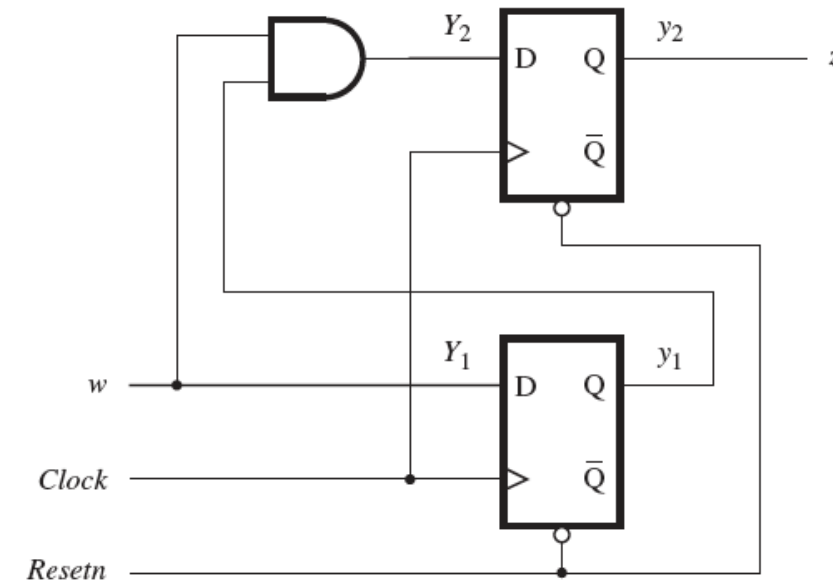


	Present state y_2y_1	Next state		Output z
		$w = 0$	$w = 1$	
		Y_2Y_1	Y_2Y_1	
A	00	00	01	0
B	01	00	10	0
C	10	00	10	1
	11	<i>dd</i>	<i>dd</i>	<i>d</i>



=

	Present state y_2y_1	Next state		Output z
		$w = 0$	$w = 1$	
		Y_2Y_1	Y_2Y_1	
A	00	00	01	0
B	01	00	11	0
C	11	00	11	1
	10	<i>dd</i>	<i>dd</i>	<i>d</i>



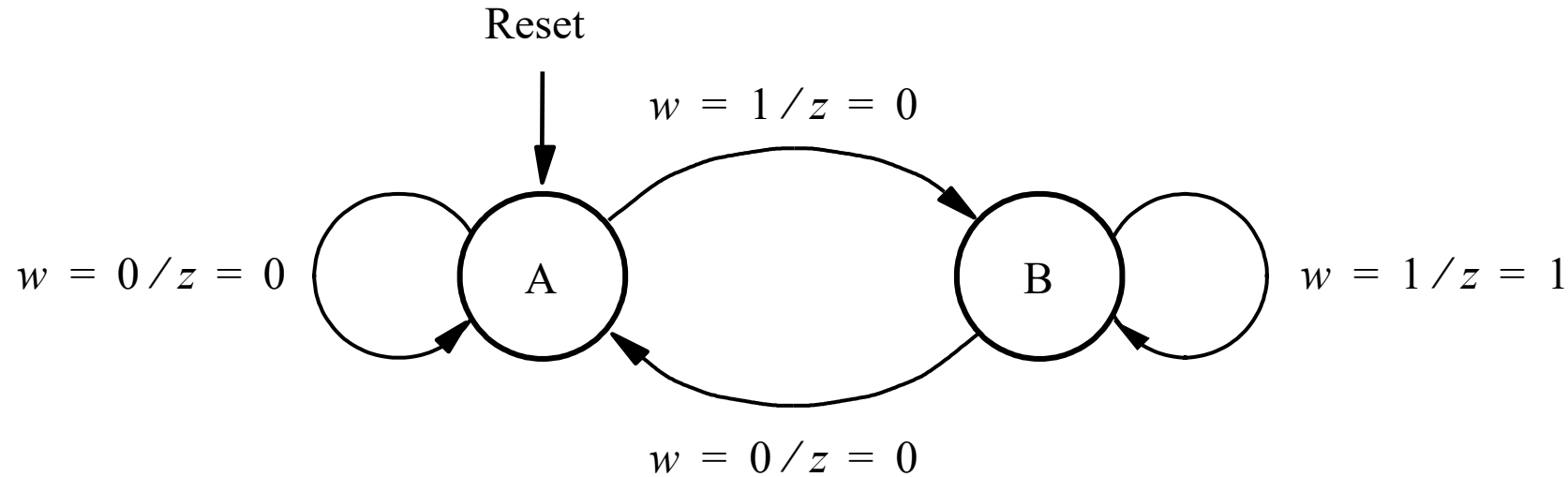
A Simple Example (Mealy)

- Circuit has one input, w , and one output, z .
- Changes occur on positive clock edge.
- z is 1 if w is 1 during last two ~~two~~ **consecutive** clock cycles.

“The specification requires that the output z should be equal to 1 in the same clock cycle when the second occurrence of $w = 1$ is detected

Clock cycle:	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}
w :	0	1	0	1	1	0	1	1	1	0	1
z :	0	0	0	0	1	0	0	1	1	0	0

Mealy State Diagram & Table



“A key point to understand is that during the present clock cycle the output value corresponds to the label on the arc emanating from the present-state node”

Present state	Next state		Outputz	
	w = 0	w = 1	w = 0	w = 1
A	A	B	0	0
B	A	B	0	1

State Assignment



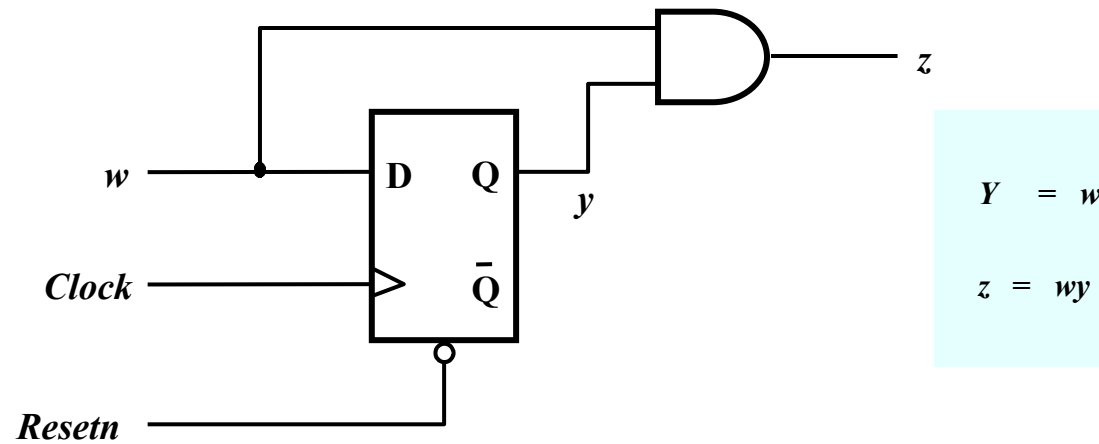
Present state	Next state		Output z	
	$w = 0$	$w = 1$	$w = 0$	$w = 1$
A	A	B	0	0
B	A	B	0	1

	Present state	Next state		Output	
		$w = 0$	$w = 1$	$w = 0$	$w = 1$
	y	Y	Y	z	z
A	0	0	1	0	0
B	1	0	1	0	1

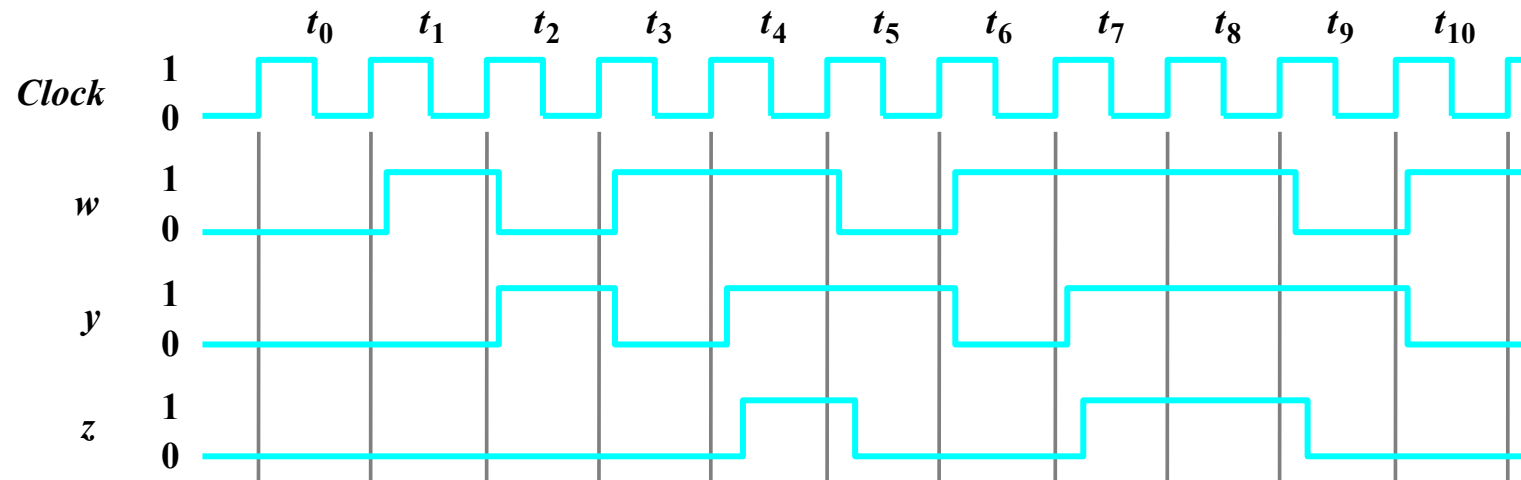
$$Y = w$$

$$z = wy$$

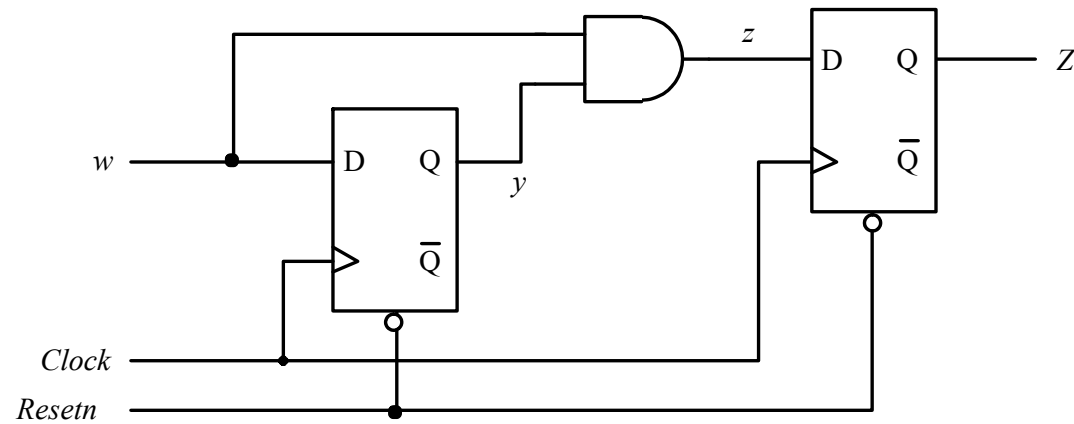
Mealy FSM Implementation



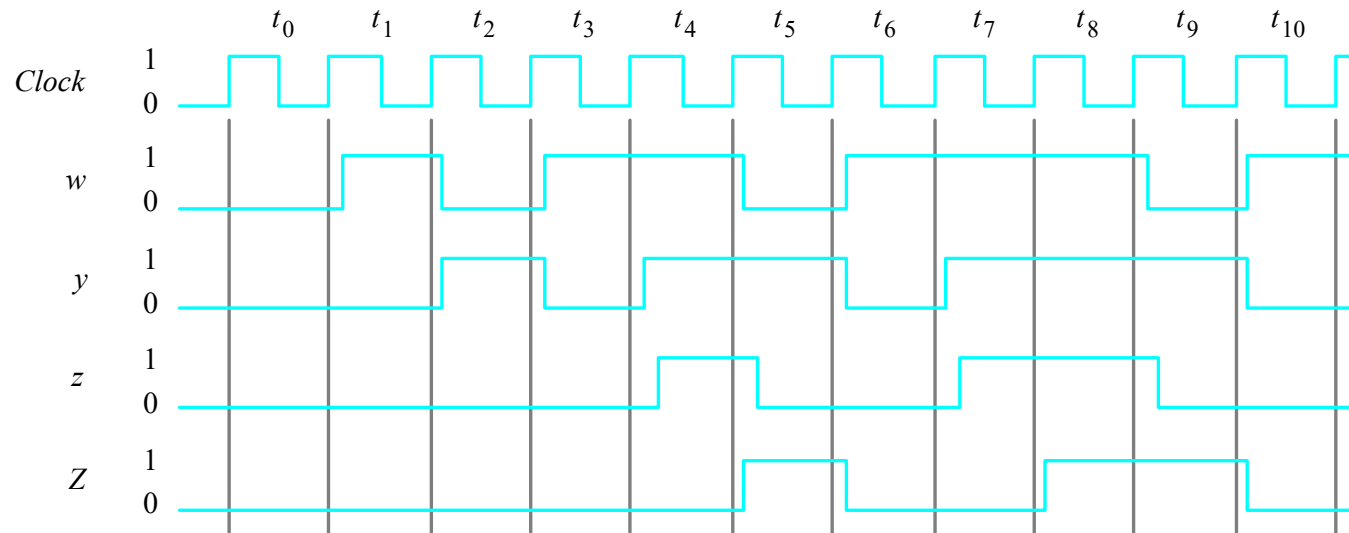
(a) Circuit



Mealy to Moore Conversion

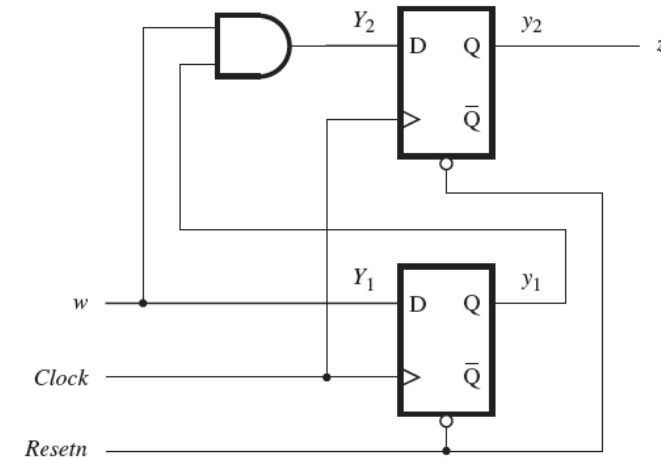


(a) Circuit



(b) Timing diagram

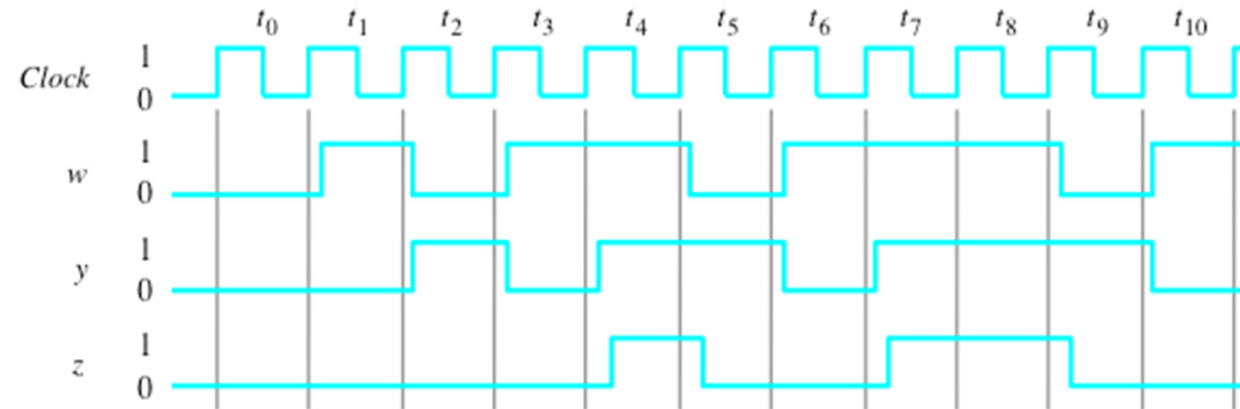
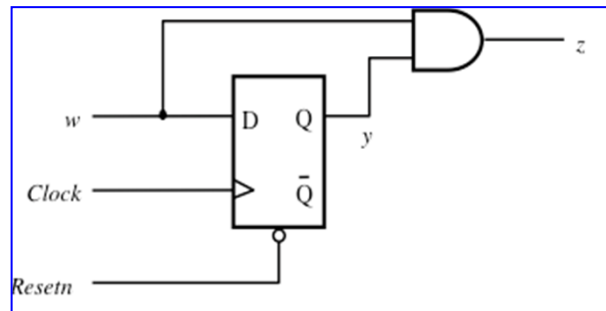
Add a flip-flop to the output



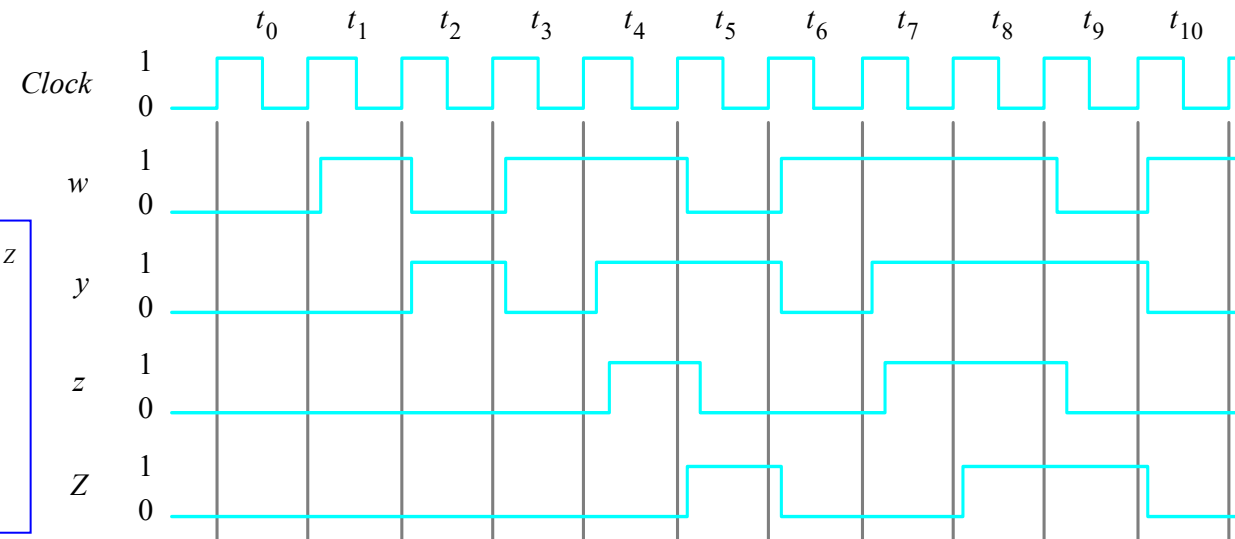
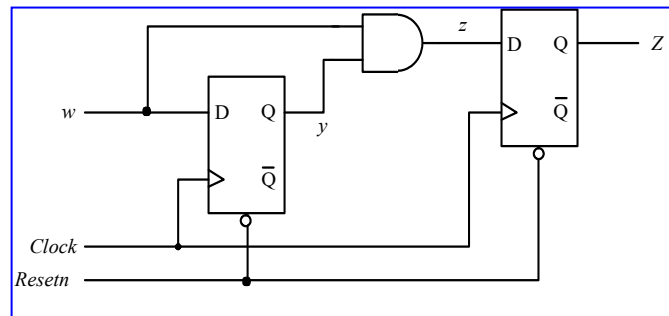
Mealy vs. Moore



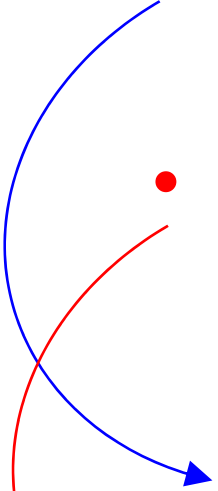
Mealy Machine:



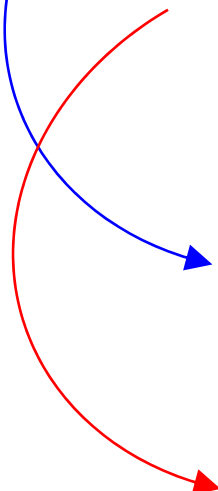
Moore Machine:



- *Moore machines* – output is determined only by present state.
- *Mealy machines* – output depends on both present state and input values.



Clockcycle:	t₀	t₁	t₂	t₃	t₄	t₅	t₆	t₇	t₈	t₉	t₁₀
w:	0	1	0	1	1	0	1	1	1	0	1
z:	0	0	0	0	0	1	0	0	1	1	0



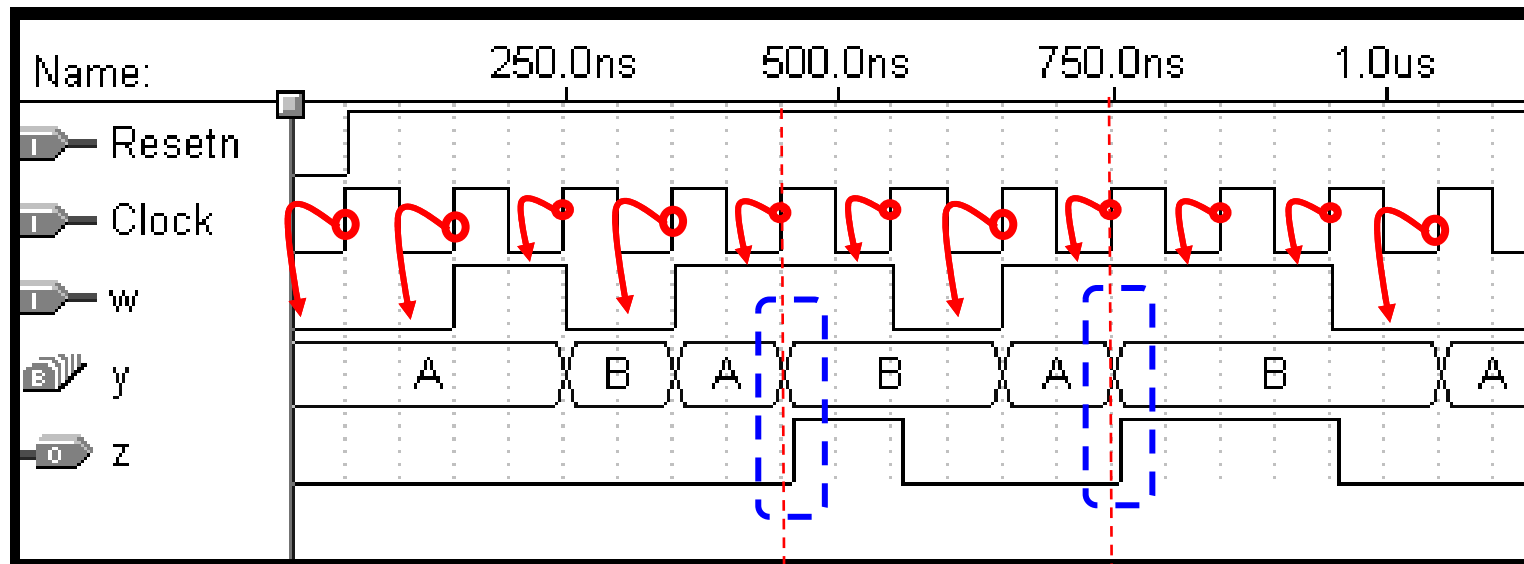
Clock cycle:	t₀	t₁	t₂	t₃	t₄	t₅	t₆	t₇	t₈	t₉	t₁₀
w:	0	1	0	1	1	0	1	1	1	0	1
z:	0	0	0	0	1	0	0	1	1	0	0

Sequences of input and output signals

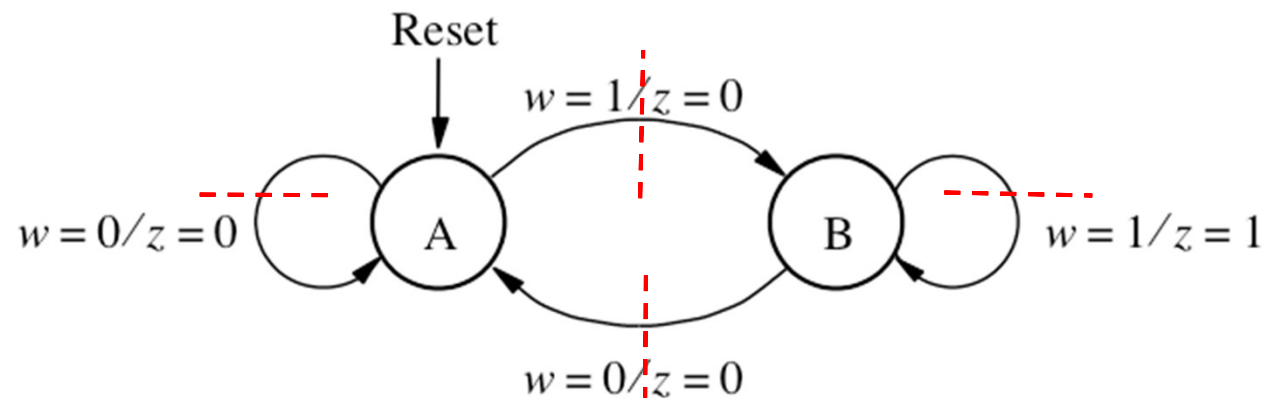
Simulation results for the Mealy machine



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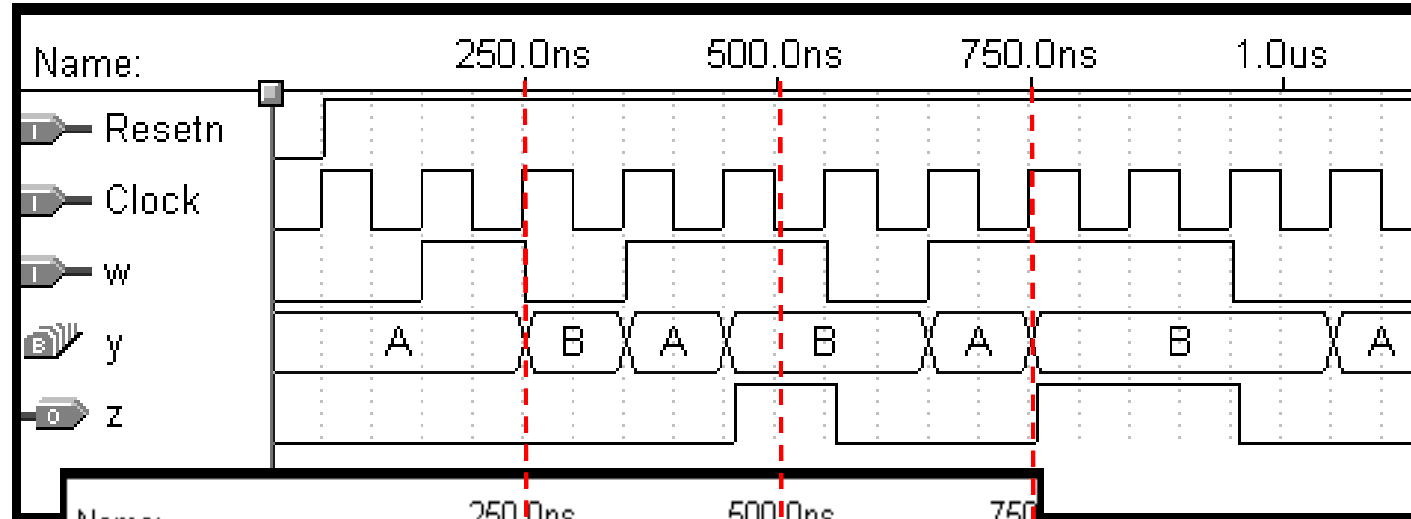


Note that all the changes in the input signal (w) occur right after the clock positive edge (after a small delay)

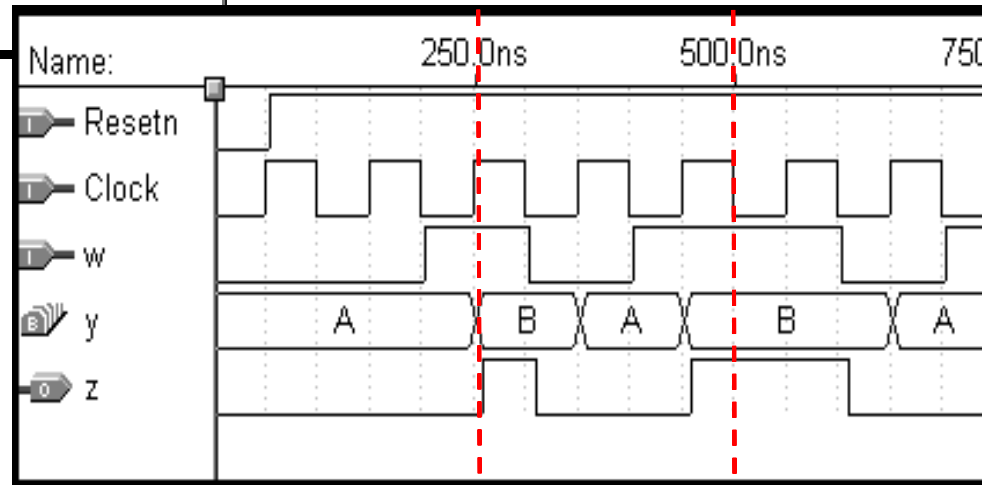


Simulation results for the Mealy machine

What if the changes in the input signal (w) occur right after the clock negative edge?

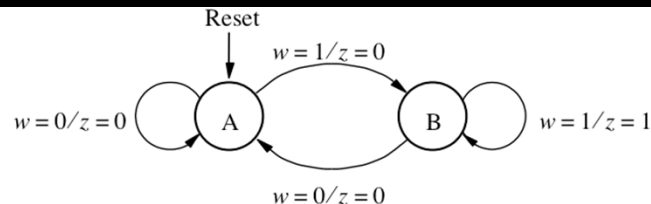


**Same input sequence
but slightly shifted
causes a glitch problem!**



**Potential problem with asynchronous inputs
to a Mealy FSM**

1. Each clock cycle 100ns
2. 1st input pulse is only 100ns long
3. You need two consecutive 1s; hence pulse has to be 200ns long to produce 1!

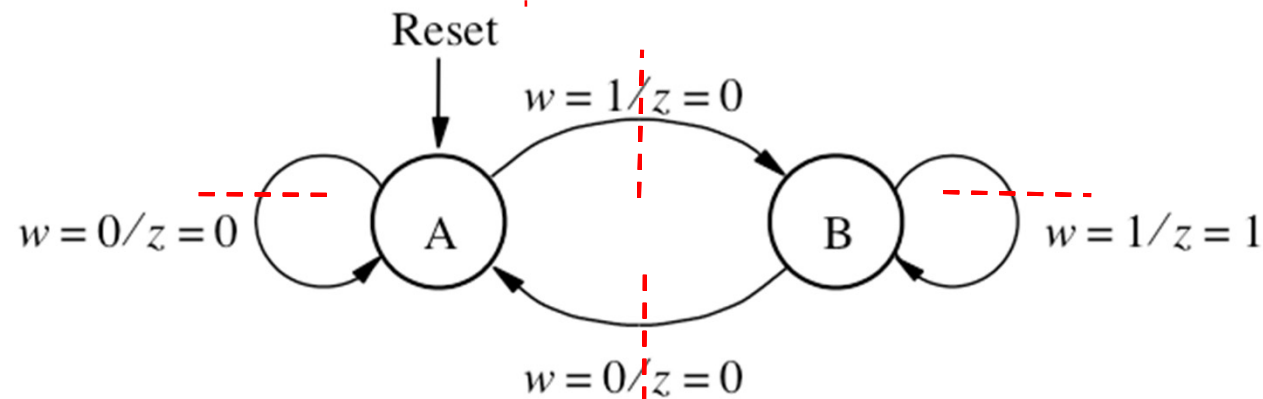
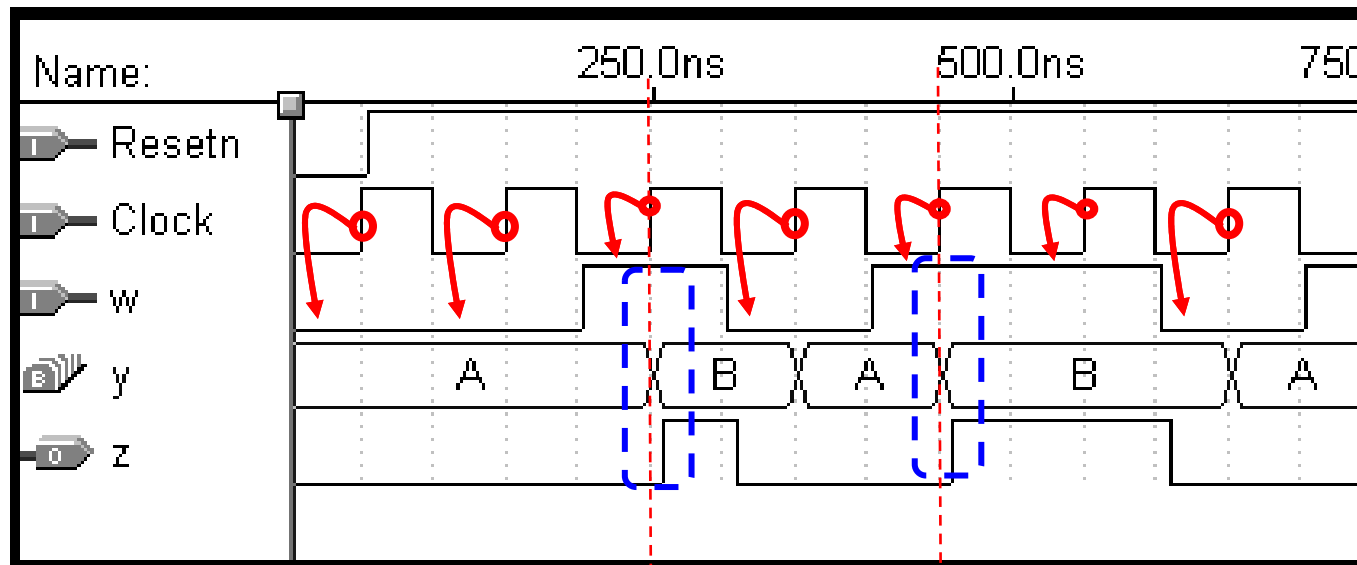


Simulation results for the Mealy machine



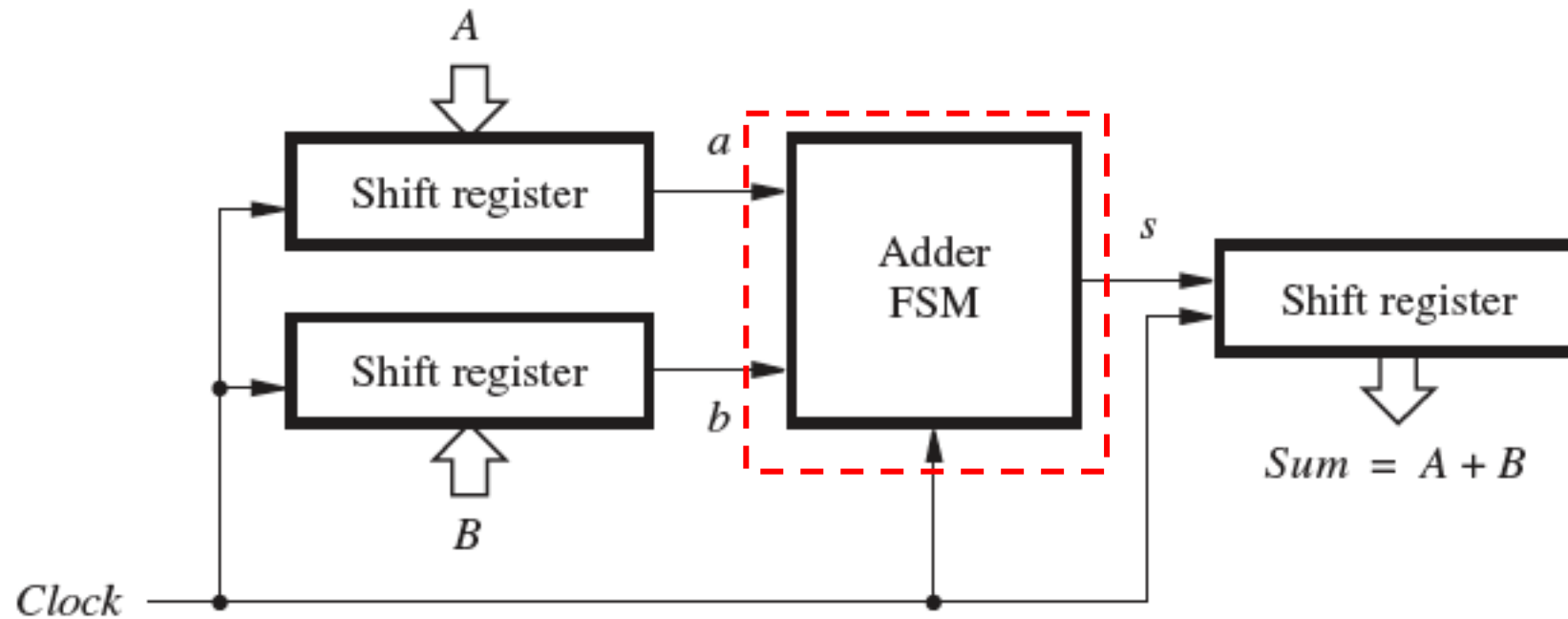
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Potential problem with asynchronous inputs to a Mealy FSM



Example: Serial Adder (Mealy)

- Draw the **Mealy** state diagram of a serial adder using minimum number of states.

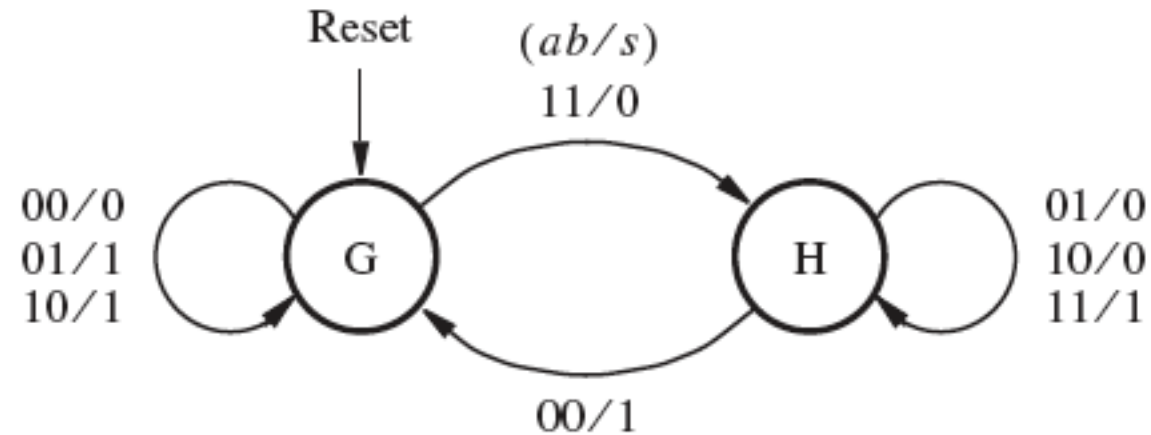


Example: Serial Adder (Mealy)

- **Assign meaning to each state.**

Example: Serial Adder (Mealy)

- Assign meaning to each state.

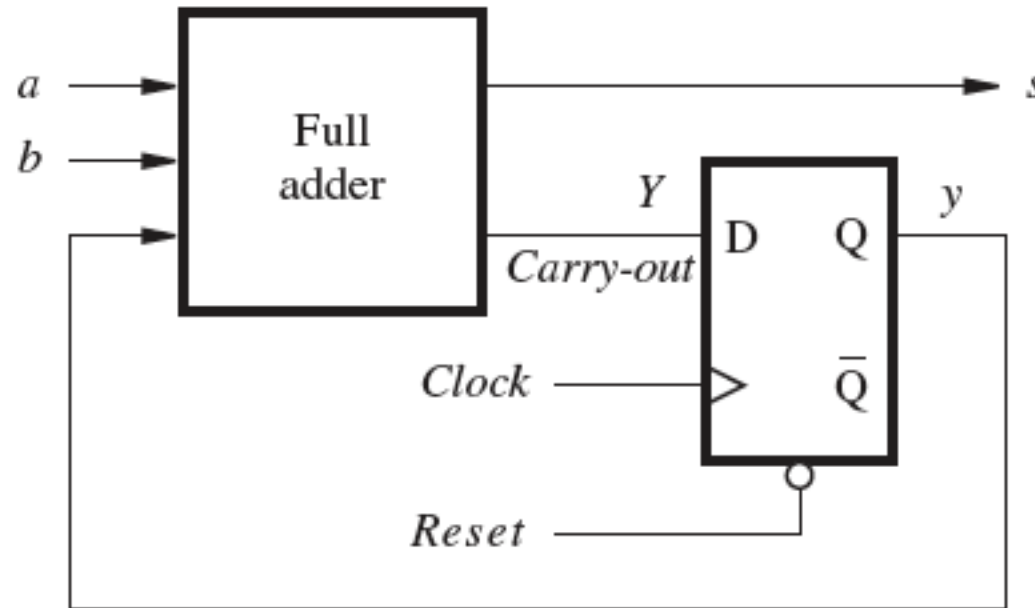


G: carry-in = 0

H: carry-in = 1

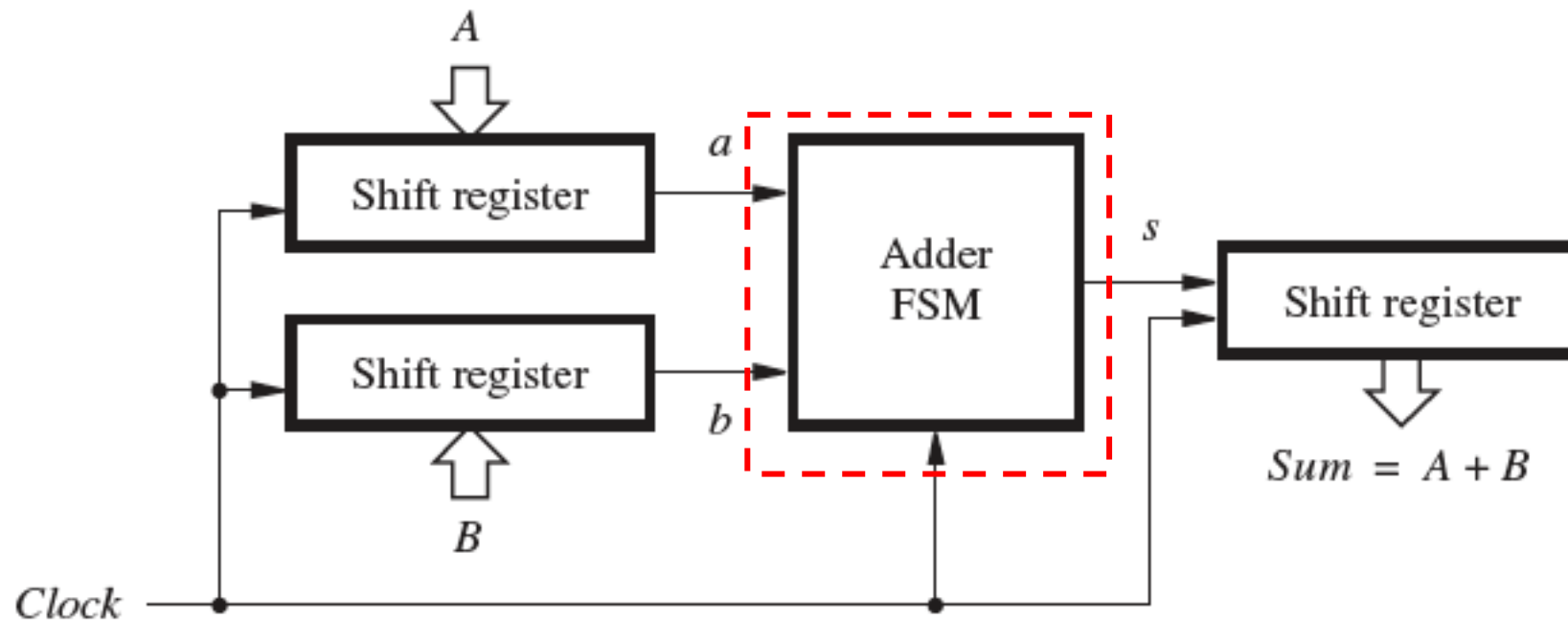
Example: Serial Adder (Mealy)

- If you solve all the way, you'll get below cct.



Example: Serial Adder (Moore)

- Draw the **Moore** state diagram of a serial adder using minimum number of states



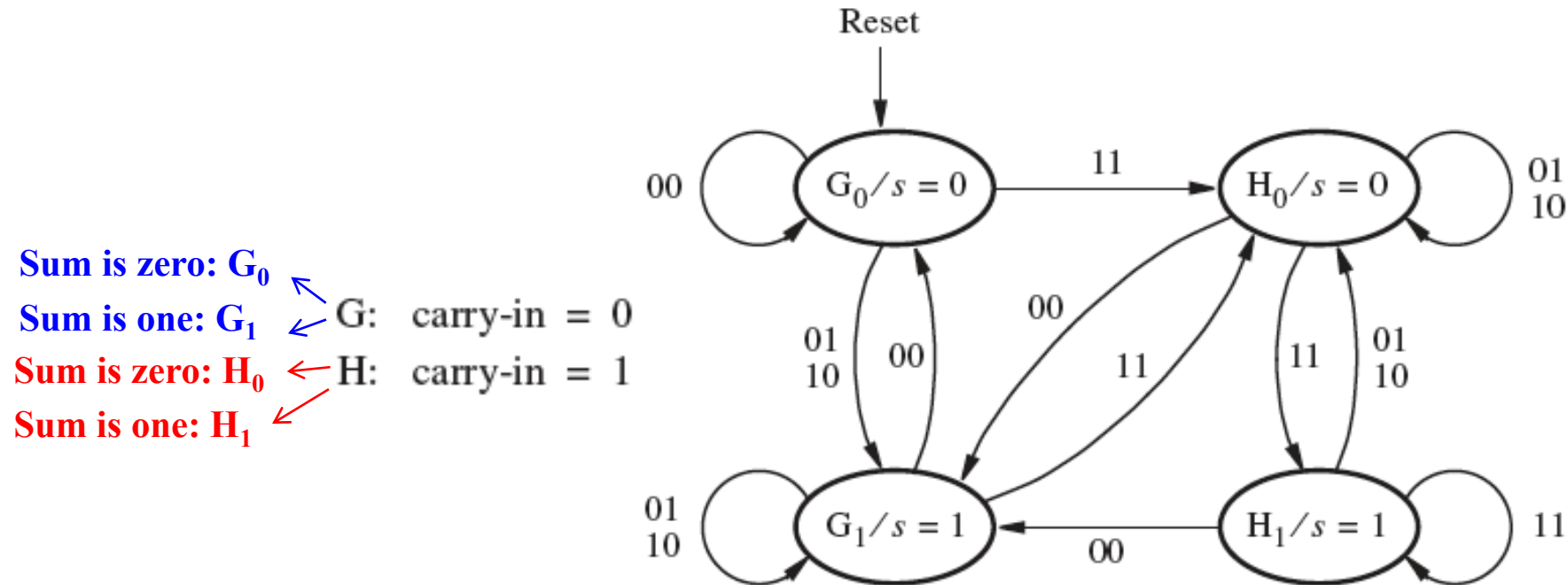
Example: Serial Adder (Moore)

- **Assign meaning to each state.**

Example: Serial Adder (Moore)

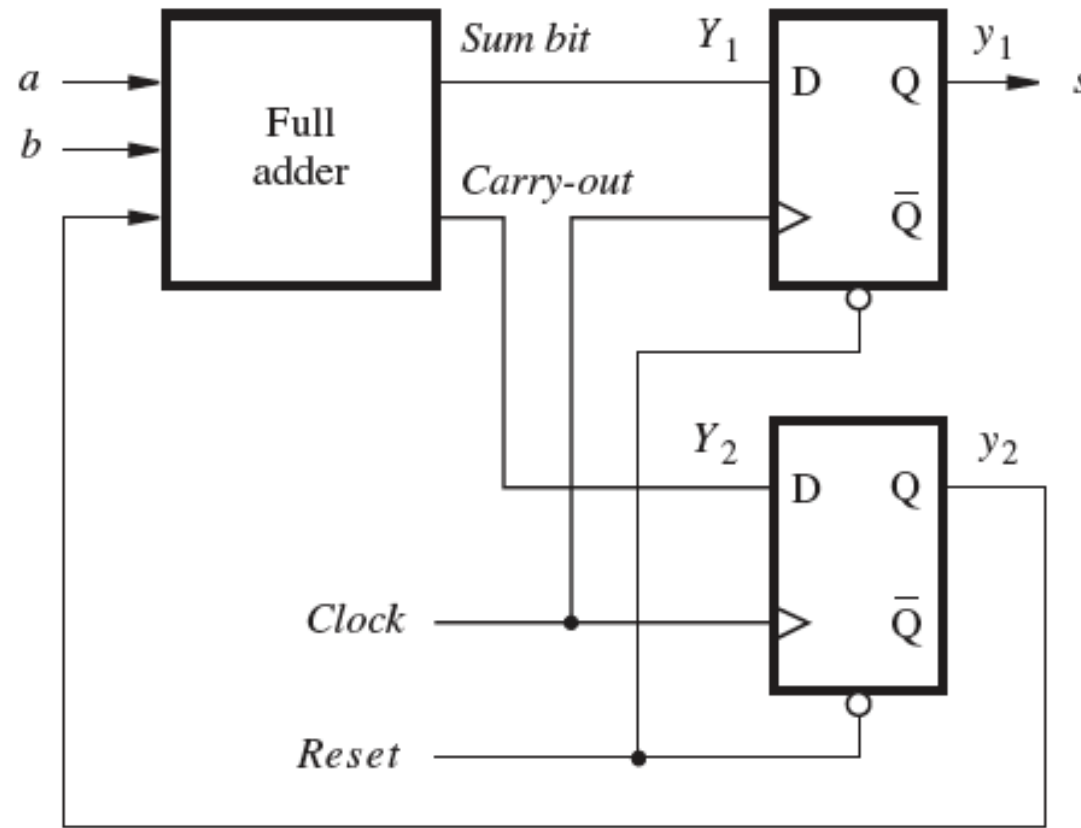


- Assign meaning to each state.



Example: Serial Adder (Moore)

- Moore logic schematic:





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Thank you for listening !

